

AWS Setup and Security Alerting System

Prepared by: Olatunji Taoheed Lawal
Cybersecurity Analyst

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Project Overview:

In this project, an AWS Free Tier account was opened, and security alerting mechanisms leveraging EventBridge, SNS (Simple Notification Service), and AWS Cloud Trail were implemented. The primary objective was to monitor and generate alerts upon detection of GetCallerIdentity API requests initiated by the AWS root user, which could indicate unauthorized access attempts or security vulnerabilities.

Objectives:

1. Creation of an AWS Free Tier account to conduct experiments in the cloud.
2. Make sure CloudTrail is set up to record AWS API activities.
3. SNS (Simple Notification Service) from Amazon can be used to deliver email notifications.
4. Configure Amazon EventBridge so that root user actions will cause alerts to be sent.
5. To verify functionality, use the AWS CLI to test the configuration.

Technologies Used:

- **AWS CloudTrail:** Tracks API calls and logs events.
- **Amazon SNS:** Sends notifications to subscribed users.
- **Amazon EventBridge:** Creates event-driven triggers.
- **Kali Linux CLI:** Used to test the security setup.

Project Steps:

1. AWS Free Tier Account Setup

- Created a Free Tier AWS account on aws.amazon.com.
- Configured billing information and identity verification.
- Selected "Basic Support – Free" for cost efficiency.

New Feature

Amazon SNS now supports High Throughput FIFO topics. [Learn more](#) 

Subscription: cd55aafb-1ee5-4aeb-b8d7-0544a52d8e1e

Details

ARN

arn:aws:sns:eu-north-1:992382659664:GetCallerAlert:cd55aafb-1ee5-4aeb-b8d7-0544a52d8e1e

Endpoint

olatunjilawal1@gmail.com

Topic

[GetCallerAlert](#)

Subscription Principal

arn:aws:iam::992382659664:root

Status

 **Confirmed**

Protocol

EMAIL

2. Configured AWS CloudTrail

- Accessed **CloudTrail** in the AWS Management Console.

- Created a new trail named MyCloudTrail.
- Enabled **Management Event Logging** for all read/write events.
- Configured logs to be stored in an **S3 bucket** (my-cloudtrail-logs-uniqueid).
- Enabled **multi-region logging** for full coverage.

Trails									
Copy events to Lake Refresh Delete Create trail									
	Name ▲	Home region ▼	Multi-region trail ▼	Insights ▼	Organization trail ▼	S3 bucket ▼	Log file prefix ▼	Cloud Watch Logs log group ▼	Status ▼
☐	alerttrail	Europe (Stockholm)	Yes	Disabled	No	aws-cloudtrail-logs-992382659664-8858971b	-	-	✔ Logging

CloudTrail

Trails

Create trail

Step 1

Choose trail attributes

Step 2

Choose log events

Step 3

Review and create

Choose log events

Events

Info

Record API activity for individual resources, or for all current and future resources in AWS account. Additional charges apply

Event type

Choose the type of events that you want to log.

☒ **Management events**
 Capture management operations performed on your AWS resources.

☐ **Data events**
 Log the resource operations performed on or within a resource.

☐ **Insights events**
 Identify unusual activity, errors, or user behavior in your account.

☐ **Network activity events**
 Network activity events provide information about resource operations performed on a resource within a virtual private cloud endpoint.

Management events

Info

Management events show information about management operations performed on resources in your AWS account.

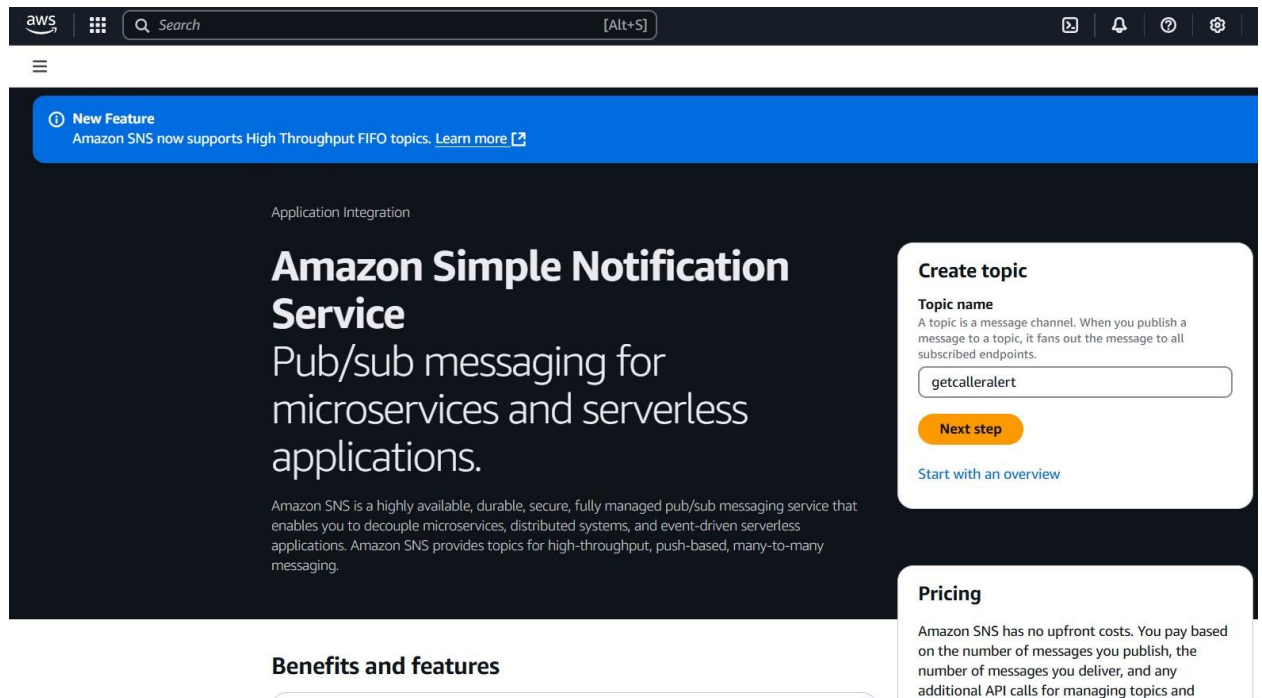
No additional charges apply to log management events on this trail because this is your first copy of management events.

API activity

3. Set up SNS for Email Alerts

- Created an **SNS topic** named APICallAlert.
- Subscribed an email address to receive notifications.

- Confirmed the email subscription via the AWS verification link.
- Linked **CloudTrail** to SNS to trigger notifications when specific events occur.



4. Configuring EventBridge for Root API Call Alerts

- Created an **EventBridge Rule** named GetCallerIdentityAlert.
- Defined a **custom event pattern** to detect GetCallerIdentity API calls from the root user:

```
{
  "source": ["aws.sts"],
  "detail-type": ["AWS API Call via CloudTrail"],
  "detail": {
    "eventSource": ["sts.amazonaws.com"],
    "eventName": ["GetCallerIdentity"],
    "userIdentity": {
      "type": ["Root"]
    }
  }
}
```

- Configured **SNS** as the target for this rule to send notifications.
- Allowed EventBridge to create an IAM role for permissions.

Identity and Access
Management (IAM)

Q Search IAM

Dashboard

▼ Access management

User groups

Users

Roles

Policies

My security credentials Root user Info

The root user has access to all AWS resources in this account, and we recommend following [best practices](#). To learn more about the types of AWS credentials and how they're used, see [AWS Security Credentials](#) in AWS General Reference

Account details

Edit account name, email, and password

Account name

ThritaGlobal

Email address

olatunji1@gmail.com

AWS account ID

992382659664

Canonical user ID

1389144d4766fb9a50271b7ce479dabbbf2d9c9dbd64f5fc2c0b4de049d09b90

5. Testing the Setup via kali Linux CLI

- Installed and configured **kali linux** with root credentials.
- Ran the following command to simulate an API call:

aws sts get-caller-identity

- Verified that the API call was logged in **CloudTrail**.
- Checked the SNS email inbox for the alert notification.

```
File Actions Edit View Help
(olatunji@customer)-[~]
$ aws sts get-caller-identity
{
  "UserId": "992382659664",
  "Account": "992382659664",
  "Arn": "arn:aws:iam::992382659664:root"
}
```

Key Takeaways

- Successfully implemented **real-time monitoring** for root user API calls.
- Demonstrated how to configure **AWS security alerting** using CloudTrail, SNS, and EventBridge.
- Gained hands-on experience with AWS **event-driven automation**.
- Ensured that **email notifications** were triggered on unauthorized root account activity.

This project demonstrates my capacity to successfully monitor cloud infrastructure and apply AWS security controls.